

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			faster reaction / higher rate at higher temperature (1) particles have more energy / move faster at higher temperature (1) award (1) for any of following • more collisions per given time • more frequent collisions • greater chance of collisions • more collisions have energy above activation energy • more successful collisions	3			3		1
	(b)			rate decreases over time (1) due to reactant particles being used up / fewer reactant particles (1)	2			2		1

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	(c)			award (1) for improvement and (1) for linked explanation e.g. ensure that the concentration of acid / mass of magnesium is kept the same (1) so that any change in results can only be as a result of changing temperature (1) use gas syringe (1) more precise / easier to read accurately (1) use of balance (1) record loss of mass more accurately than volume of gas (using this apparatus) (1) make repeat measurements (1) calculate mean values which are more accurate (1) accept other sensible answers			2	2		2

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	(d)	(i)		0.0185 mol of magnesium (1) 1:1 ratio / moles hydrogen produced also 0.0185 mol of hydrogen (1) 0.037 g of hydrogen (1) accept greater number of sig figs ecf possible		3		3	3	
		(ii)		0.0185×24 (1) 0.444 dm^3 (1) accept greater number of sig figs ecf possible from (d)(i)		2		2	2	
				Question 8 total	5	5	2	12	5	4